1. A transformer works on the principle of:

(A) Mutual electromagnetic induction
(B) Self-induction only
(C) Electrostatic attraction
(D) Lorentz force

2. In an ideal transformer, the core permeability is assumed to be:

(A) Infinite
(B) Zero
(C) Unity
(D) 100

3. The core of a transformer is laminated to minimize:

(A) Eddy current loss
(B) Hysteresis loss
(C) Copper loss
(D) Friction loss

4. Hysteresis loss in a transformer core is reduced by using:

(A) Silicon steel (high silicon content)
(B) Soft copper
(C) Cast iron
(D) Aluminum

5. The EMF equation of a single-phase transformer is:

(A) E = 4.44 * f * N * Phi_m
(B) E = 4 * f * N * Phi_m
(C) E = 2.22 * f * N * Phi_m
(D) E = f * N * Phi_m

6. A step-up transformer increases voltage while:

(A) Decreasing current proportionally
(B) Increasing current proportionally
(C) Increasing frequency
(D) Decreasing power

7. The frequency of secondary voltage in a 50 Hz transformer is:

(A) 50 Hz
(B) 100 Hz
(C) 25 Hz
(D) 0 Hz (DC)

8. The transformation ratio K of a transformer is defined as:

(A) V2 / V1 = N2 / N1 = I1 / I2
(B) V1 / V2 = N2 / N1
(C) V2 / V1 = I2 / I1
(D) N1 / N2

9. In an Open-Circuit (OC) test of a transformer, rated voltage is applied to determine:

(A) Core (iron) losses
(B) Full-load copper loss
(C) Winding resistance
(D) Leakage reactance

10. In a Short-Circuit (SC) test of a transformer, rated current is circulated to determine:

(A) Full-load copper losses and equivalent impedance
(B) Core loss
(C) Magnetizing reactance
(D) Hysteresis loss only

11. Transformer core losses (iron losses) depend primarily on:

(A) Supply voltage and frequency
(B) Load current magnitude
(C) Load power factor
(D) Ambient humidity

12. Copper losses in transformer windings vary with load current I as:

(A) Proportional to I^2
(B) Proportional to I
(C) Independent of I
(D) Proportional to sqrt(I)

13. A transformer attains its maximum operating efficiency when:

(A) Copper loss equals Iron loss
(B) Copper loss is zero
(C) Iron loss is zero
(D) Load is 100% inductive

14. The primary function of transformer oil (mineral oil) in oil-immersed transformers is:

(A) Insulation and cooling
(B) Lubrication
(C) Reducing weight
(D) Preventing rust only

15. A Buchholz relay is a gas-actuated protective device installed between:

(A) Transformer tank and conservator
(B) Primary and secondary bushings
(C) Breather and tank
(D) Radiator tubes

16. The silica gel in a transformer breather is used to:

(A) Absorb moisture from air entering the conservator
(B) Filter dust particles only
(C) Cool the oil
(D) Dissolve combustible gases

17. Fresh, dry silica gel is blue in color; upon absorbing moisture it turns:

(A) Pink / Pale pink
(B) Dark green
(C) Yellow
(D) Black

18. An autotransformer has:

(A) A single continuous winding serving as both primary and secondary
(B) Two isolated windings
(C) Three isolated windings
(D) Zero windings

19. Compared to a conventional two-winding transformer of equal rating, an autotransformer requires:

(A) Less copper conductor material
(B) More copper conductor material
(C) Identical copper material
(D) No core

20. The function of a commutator in a DC generator is to convert:

(A) AC armature voltage into DC output voltage (mechanical rectifier)
(B) DC into AC
(C) Mechanical power to thermal power
(D) Speed into torque

21. The commutator segments of a DC machine are insulated from one another by thin sheets of:

(A) Mica
(B) Paper
(C) PVC
(D) Asbestos

22. The brushes of a DC machine are universally fabricated from:

(A) Carbon / Graphite
(B) Copper
(C) Aluminum
(D) Cast steel

23. The EMF equation of a DC generator with P poles, Phi flux, Z conductors, and N rpm is:

(A) Eg = (Phi * Z * N * P) / (60 * A)
(B) Eg = (Phi * Z * N) / (60 * P)
(C) Eg = (P * Z * N) / 60
(D) Eg = Phi * N * A

24. For a simplex lap-wound DC machine, the number of parallel paths A is equal to:

(A) Number of poles P
(B) 2
(C) 4
(D) P / 2

25. For a simplex wave-wound DC machine, the number of parallel paths A is always equal to:

(A) 2
(B) P
(C) P * 2
(D) 4

26. Lap winding is preferred for DC machines designed for:

(A) High current and low voltage
(B) High voltage and low current
(C) Ultra-high speed only
(D) Zero current

27. Wave winding is preferred for DC machines designed for:

(A) High voltage and low current
(B) High current and low voltage
(C) Low torque
(D) Zero flux

28. In a DC motor, the back-EMF Eb develops according to:

(A) Lenz's Law, opposing applied terminal voltage V
(B) Ampere's Law aiding V
(C) Coulomb's Law
(D) Ohm's Law only

29. The armature current Ia of a DC motor is given by:

(A) Ia = (V - Eb) / Ra
(B) Ia = (V + Eb) / Ra
(C) Ia = V / Ra
(D) Ia = Eb / Ra

30. The electromagnetic torque developed by a DC motor is proportional to:

(A) Phi * Ia (Flux * Armature Current)
(B) V * Ia
(C) Eb / Ia
(D) Phi / Ia

31. A DC series motor must NEVER be started under:

(A) No-load (zero mechanical load)
(B) Full-load
(C) Half-load
(D) Heavy load

32. Under no-load conditions, a DC series motor develops:

(A) Dangerously high runaway speed
(B) Zero speed (stalls)
(C) Rated speed
(D) Negative rotation

33. Which DC motor provides nearly constant speed across all load variations?

(A) DC Shunt Motor
(B) DC Series Motor
(C) Differential Compound Motor
(D) Universal Motor

34. Which DC motor develops the highest starting torque and is used for electric traction and cranes?

(A) DC Series Motor
(B) DC Shunt Motor
(C) Cumulative Compound Motor
(D) Permanent Magnet Motor

35. A 3-phase induction motor operates on the principle of:

(A) A rotating magnetic field (RMF) inducing rotor currents
(B) Direct conduction to rotor
(C) Electrostatic induction
(D) Reluctance torque only

36. The synchronous speed Ns of a 3-phase induction motor with P poles at frequency f is:

(A) Ns = (120 * f) / P
(B) Ns = (60 * f) / P
(C) Ns = (120 * P) / f
(D) Ns = f / (120 * P)

37. For a 4-pole 3-phase induction motor operating on a 50 Hz supply, synchronous speed is:

(A) 1500 rpm
(B) 3000 rpm
(C) 1000 rpm
(D) 750 rpm

38. The slip 's' of an induction motor running at rotor speed N is defined as:

(A) (Ns - N) / Ns
(B) (N - Ns) / N
(C) N / Ns
(D) Ns / N

39. At standstill (rotor stationary, N = 0), the slip of an induction motor is:

(A) 1.0 (100%)
(B) 0
(C) 0.5
(D) Infinite

40. At synchronous speed (N = Ns), the slip of an induction motor is:

(A) 0
(B) 1.0
(C) -1.0
(D) 0.04

41. The frequency of rotor induced EMF (fr) in an induction motor running at slip s is:

(A) fr = s * f
(B) fr = f / s
(C) fr = f
(D) fr = (1 - s) * f

42. In a squirrel-cage induction motor, rotor bars are slightly skewed to:

(A) Reduce cogging, magnetic crawling, and acoustic humming
(B) Increase rotor resistance
(C) Improve ventilation
(D) Reduce cost

43. The starting torque of a 3-phase slip-ring induction motor can be increased by:

(A) Adding external resistance to the rotor circuit via slip rings
(B) Reducing supply voltage
(C) Shorting stator coils
(D) Increasing air gap

44. In a 3-phase induction motor, maximum torque (breakdown torque) occurs at slip sm equal to:

(A) R2 / X2
(B) X2 / R2
(C) R2 * X2
(D) sqrt(R2 / X2)

45. The magnitude of maximum torque (T_max) in a 3-phase induction motor is:

(A) Independent of rotor resistance R2
(B) Directly proportional to R2
(C) Inversely proportional to R2
(D) Zero

46. Adding external resistance to the rotor circuit of a slip-ring motor shifts T_max to:

(A) Higher slip values without changing the magnitude of T_max
(B) Lower slip values
(C) Zero slip
(D) Higher T_max value

47. A 4-pole, 50 Hz induction motor runs at 1440 rpm. Its operating slip is:

(A) 0.04 (4%)
(B) 0.02 (2%)
(C) 0.06 (6%)
(D) 0.05 (5%)

48. For the above motor (s = 0.04, f = 50 Hz), the rotor current frequency is:

(A) 2.0 Hz
(B) 50 Hz
(C) 0.04 Hz
(D) 100 Hz

49. Crawling in a 3-phase induction motor is predominantly caused by the:

(A) 7th space harmonic of the rotating magnetic field
(B) 5th harmonic
(C) 3rd harmonic
(D) 11th harmonic

50. Cogging (magnetic locking) in an induction motor occurs when:

(A) Number of stator slots equals or is an integer multiple of rotor slots
(B) Supply voltage drops by 10%
(C) Load torque is high
(D) Rotor bars are skewed

51. The speed of a DC shunt motor above rated speed is achieved by:

(A) Flux weakening (inserting rheostat in shunt field circuit)
(B) Armature resistance control
(C) Reducing supply voltage
(D) Reversing armature leads

52. Ward-Leonard speed control of a DC motor provides smooth, bidirectional speed control by:

(A) Varying the armature voltage using a dedicated motor-generator set
(B) Switching field poles
(C) Using series rheostats
(D) Tapping transformer

53. Armature reaction in a DC machine refers to the distorting and demagnetizing effect of:

(A) Armature magnetic flux on main field flux
(B) Main field on armature core
(C) Commutator sparking
(D) Eddy currents

54. Armature reaction in a DC generator shifts the Magnetic Neutral Axis (MNA):

(A) In the direction of rotation
(B) Opposite to the direction of rotation
(C) By strictly 90 degrees
(D) It does not shift

55. Interpoles (commutating poles) in DC machines are connected in:

(A) Series with the armature winding to neutralize cross-magnetization in commutating zone
(B) Parallel with field
(C) Parallel with armature
(D) Series with shunt field

56. Compensating windings in heavy-duty DC machines are placed in:

(A) Pole shoes in series with armature to neutralize armature reaction under pole faces
(B) Armature slots
(C) Yoke
(D) Commutator

57. In a transformer, the per-unit impedance referred to primary is:

(A) Identical to per-unit impedance referred to secondary
(B) K^2 times secondary
(C) 1/K^2 times secondary
(D) Zero

58. All-Day Efficiency of a distribution transformer is calculated on the basis of:

(A) Energy output in kWh over 24 hours / Energy input in kWh over 24 hours
(B) Power in kW at full load
(C) Core losses only
(D) Maximum efficiency load

59. Distribution transformers are designed to have maximum efficiency at approximately:

(A) 50% to 70% of full load (since they operate on light load most of the day)
(B) 100% full load
(C) 120% overload
(D) 10% load

60. Power transformers installed in generating stations are designed to achieve maximum efficiency at:

(A) Near 100% full rated load
(B) 50% load
(C) 25% load
(D) No load

61. In a synchronous alternator, the stator winding is called the:

(A) Armature winding
(B) Field winding
(C) Damper winding
(D) Compensating winding

62. Modern large commercial alternators use stationary armatures and rotating fields because:

(A) It is easier to insulate high-voltage windings and collect high power directly without slip rings
(B) It reduces weight
(C) Permanent magnets are cheaper
(D) It avoids all losses

63. The pitch factor (chording factor) Kp of a fractional-pitch alternator coil with chording angle alpha is:

(A) cos(alpha / 2)
(B) sin(alpha / 2)
(C) cos(alpha)
(D) sin(alpha)

64. The distribution factor (breadth factor) Kd for m slots per pole per phase with slot pitch beta is:

(A) sin(m * beta / 2) / [m * sin(beta / 2)]
(B) m * sin(beta / 2)
(C) cos(m * beta / 2)
(D) m * beta

65. Voltage regulation of an alternator supplying a leading power factor load can be:

(A) Negative (terminal voltage rises on load)
(B) Always positive
(C) Zero only
(D) Infinite

66. The synchronous impedance Zs of an alternator is defined as:

(A) Open-circuit terminal voltage / Short-circuit armature current at identical field excitation
(B) V_rated / I_rated
(C) Ra + j*Xl
(D) Ra only

67. Synchronous motors are NOT self-starting because:

(A) The rotor has high mechanical inertia and cannot accelerate to synchronous speed in a half-cycle
(B) They lack magnetic fields
(C) They require DC stator excitation
(D) They have no armature

68. Damper windings (amortisseur windings) on synchronous motor rotor pole faces serve to:

(A) Start the motor as an induction motor and suppress hunting
(B) Increase field current
(C) Produce DC power
(D) Improve insulation

69. An over-excited synchronous motor operating with zero mechanical load is called a:

(A) Synchronous condenser (acts as a large power-factor correcting capacitor)
(B) Universal motor
(C) Dynamometer
(D) Amplidyne

70. The V-curves of a synchronous motor plot:

(A) Armature current Ia versus Field current If for constant mechanical load
(B) Power factor versus speed
(C) Torque versus slip
(D) Voltage versus speed

71. The Inverted V-curves of a synchronous motor plot:

(A) Power factor (cos phi) versus Field current If
(B) Ia versus If
(C) Torque versus speed
(D) Slip versus voltage

72. A universal motor is a series motor designed to operate efficiently on:

(A) Both AC and DC power supplies
(B) DC only
(C) 3-phase AC only
(D) High-frequency radio waves

73. Stepper motors are brushless DC motors that convert digital input pulses into:

(A) Discrete, precise angular mechanical increments (steps)
(B) Continuous uncontrolled rotation
(C) Pure linear motion
(D) Variable frequency AC

74. A brushless DC (BLDC) motor replaces the mechanical commutator and brushes with:

(A) Electronic switching inverter driven by Hall effect position sensors
(B) Centrifugal switches
(C) Slip rings
(D) Carbon blocks

75. The core of an induction motor stator is constructed from stacked laminations of:

(A) Cold-Rolled Grain-Oriented (CRGO) or non-oriented silicon steel
(B) Pure copper
(C) Cast iron
(D) Wood

76. In Blondel's two-reaction theory for salient-pole synchronous machines, armature MMF is resolved into:

(A) Direct-axis (d-axis) and Quadrature-axis (q-axis) components
(B) Sequence components
(C) Harmonic components
(D) Stator and rotor coordinates

77. In a salient-pole alternator, direct-axis synchronous reactance Xd is:

(A) Greater than quadrature-axis synchronous reactance Xq (Xd > Xq)
(B) Equal to Xq
(C) Smaller than Xq (Xd < Xq)
(D) Zero

78. The reluctance power developed by a salient-pole synchronous machine without field excitation is proportional to:

(A) (Xd - Xq) * sin(2 * delta)
(B) (Xd + Xq) * sin(delta)
(C) Xd * Xq * cos(delta)
(D) sin(delta)

79. In Potier's method (Zero Power Factor test) of an alternator, the triangle constructed determines:

(A) Armature leakage reactance Xl and armature reaction MMF Fa
(B) Synchronous reactance only
(C) Core loss
(D) Winding resistance

80. Hunting in synchronous machines refers to:

(A) Low-frequency mechanical oscillations of the rotor around synchronous equilibrium following disturbance
(B) Speed runaway
(C) Loss of field
(D) Complete bearing seizure

81. The natural oscillation frequency of a synchronous machine during hunting is given by:

(A) fn = (1 / 2*pi) * sqrt(Ps / M), where Ps is synchronizing power coefficient and M is inertia
(B) fn = 50 Hz
(C) fn = 120 * f / P
(D) fn = 0

82. In the exact equivalent circuit of a 3-phase induction motor, mechanical power developed P_mech is modeled by resistance:

(A) R2' * [(1 - s) / s]
(B) R2' / s
(C) R2' * s
(D) R2' / (1 - s)

83. Rotor copper loss P_cu in an induction motor is related to total air-gap power P_ag by:

(A) P_cu = s * P_ag
(B) P_cu = (1 - s) * P_ag
(C) P_cu = P_ag / s
(D) P_cu = P_ag * (1 - s) / s

84. Internal gross mechanical power developed P_mech is related to air-gap power P_ag by:

(A) P_mech = (1 - s) * P_ag
(B) P_mech = s * P_ag
(C) P_mech = P_ag / (1 - s)
(D) P_mech = P_ag

85. The ratio P_ag : P_cu : P_mech in an induction motor is strictly equal to:

(A) 1 : s : (1 - s)
(B) s : 1 : (1 - s)
(C) (1 - s) : s : 1
(D) 1 : (1 - s) : s

86. During a 3-phase symmetrical short-circuit at alternator terminals, armature current exhibits:

(A) Subtransient period (X''d), Transient period (X'd), and Steady-state period (Xd)
(B) Instantaneous drop to zero
(C) Steady sinusoidal current only
(D) No DC offset

87. The subtransient reactance X''d is much smaller than steady-state Xd because:

(A) Damper windings and rotor body trap flux and prevent it from penetrating rotor iron
(B) Armature resistance drops
(C) Field current becomes zero
(D) Air gap collapses

88. In a Scott-connected transformer bank converting 3-phase to 2-phase, the teaser transformer has a tapping at:

(A) 86.6% (sqrt(3)/2) of main transformer turns
(B) 50% of turns
(C) 57.7% of turns
(D) 100% of turns

89. The main transformer in a Scott connection has a center tapping at:

(A) 50% on the 3-phase primary winding
(B) 86.6%
(C) 33.3%
(D) 75%

90. In tertiary winding transformers, the delta-connected tertiary winding serves to:

(A) Circulate 3rd harmonic currents and stabilize neutral potential
(B) Step up voltage to 1000 kV
(C) Provide cooling
(D) Eliminate copper losses

91. The magnetizing current waveform of a transformer energized with sinusoidal voltage is:

(A) Peaked and non-sinusoidal due to core magnetic saturation
(B) Purely sinusoidal
(C) Square wave
(D) DC pulse

92. The induced EMF in a transformer core contains which dominant harmonic if flux is forced to be non-sinusoidal?

(A) 3rd harmonic (150 Hz in 50 Hz systems)
(B) 2nd harmonic
(C) 4th harmonic
(D) 6th harmonic

93. In a DC machine, flashover across commutator segments is initiated by:

(A) Severe distortion of air-gap flux density by heavy armature cross-magnetization
(B) Low speed
(C) Pure copper brushes
(D) Excessive grease

94. Dynamic braking of a DC shunt motor is achieved by:

(A) Disconnecting armature from supply and reconnecting it across a braking resistor while maintaining field excitation
(B) Reversing field leads
(C) Plugging
(D) Shorting stator

95. Plugging (counter-current braking) of a DC motor involves:

(A) Reversing armature terminal polarities while inserting external current-limiting resistance
(B) Opening field
(C) Shorting terminals
(D) Increasing field voltage

96. Regenerative braking is possible in electric locomotives when:

(A) Motor back-EMF exceeds line voltage (e.g., descending a gradient)
(B) Motor speed drops to zero
(C) Supply power fails
(D) Brakes are locked mechanically

97. In vector control (Field-Oriented Control, FOC) of AC induction motors, stator current is decoupled into:

(A) Direct flux-producing current (ids) and quadrature torque-producing current (iqs)
(B) Phase A and B currents
(C) Sequence currents
(D) Harmonic components

98. FOC transforms 3-phase stationary variables (abc) into synchronous rotating reference frame (dq0) using:

(A) Clarke and Park transformations
(B) Fourier transform
(C) Laplace transform
(D) Wavelet transform

99. A doubly-fed induction generator (DFIG) used in wind turbines has its rotor connected to the grid via:

(A) Back-to-back bidirectional PWM converters (AC-DC-AC)
(B) Direct fixed connection
(C) Slip rings to resistors only
(D) Permanent magnets

100. The power rating of rotor-side converter in a DFIG operating within +-30% slip range is only:

(A) Approximately 25% to 30% of total machine rating
(B) 100% of machine rating
(C) 50%
(D) 5%

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Derive the EMF equation of a single-phase transformer, and explain the physical significance of mutual magnetic flux.

Q2. Develop the complete equivalent circuit model of a practical transformer, explaining how winding resistances and leakage reactances are referred between sides.

Q3. Explain Open-Circuit (OC) and Short-Circuit (SC) tests on a single-phase transformer, detailing how equivalent circuit parameters are extracted.

Q4. Derive the mathematical expression for Voltage Regulation of a transformer, analyzing lagging, leading, and unity power factor conditions.

Q5. Derive the condition for Maximum Efficiency in a power transformer, and contrast power transformer design with distribution transformer design.

Q6. Explain the construction, working principle, and copper saving of an Autotransformer compared to a conventional two-winding transformer.

Q7. Derive the generated EMF and developed Torque equations of a DC Machine from Lorentz Force and Faraday's Law.

Q8. Explain Armature Reaction in DC machines, its distorting and demagnetizing effects, and methods used for neutralization.

Q9. Compare DC Shunt, Series, and Compound Motors regarding torque-speed characteristics and industrial applications.

Q10. Explain Speed Control methods for DC Shunt Motors (Armature Resistance, Field Weakening, and Ward-Leonard System).

Q11. Explain the Rotating Magnetic Field (RMF) production in a 3-Phase Induction Motor, proving mathematically its constant magnitude and synchronous speed.

Q12. Derive the Torque-Slip Equation of a 3-phase Induction Motor, and analyze operating regions (motoring, generating, braking).

Q13. Explain the Synchronous Motor working principle, why it is not self-starting, and methods used to start it.

Q14. Explain V-Curves and Inverted V-Curves of Synchronous Motors, detailing the operation of a Synchronous Condenser.

Q15. Explain the Scott Connection (T-Connection) of transformers for 3-Phase to 2-Phase conversion, deriving turns ratios.

Q16. Analyze the Equivalent Circuit and Power Flow Diagram of a 3-Phase Induction Motor, deriving power relations.

Q17. Explain the phenomena of Cogging and Crawling in 3-Phase Induction Motors and how they are eliminated.

Q18. Analyze Potier's Method (Zero Power Factor Method) for determining voltage regulation of an alternator.

Q19. Explain Blondel's Two-Reaction Theory for Salient-Pole Synchronous Machines, deriving the power-angle relation.

Q20. Explain Field-Oriented Control (Vector Control) of 3-Phase Induction Motors, covering Clarke and Park transformations.

Q21. Explain Doubly-Fed Induction Generators (DFIG) in wind energy conversion systems, detailing sub-synchronous and super-synchronous operation.

Q22. Explain Transformer Inrush Current, deriving the mathematical formula for flux doubling and analyzing protection measures.

Q23. Explain Stepper Motors: Variable Reluctance, Permanent Magnet, and Hybrid types, deriving step angle.

Q24. Explain Brushless DC (BLDC) Motors, including electronic commutation, Hall sensors, and inverter topologies.

Q25. Analyze Electrical Machine Losses and derive the separation of Eddy Current and Hysteresis losses from experimental data.

Q26. Explain Switched Reluctance Motors (SRM), covering torque production, stator/rotor pole configurations, and converters.

Q27. Analyze the Starting Methods of 3-Phase Squirrel-Cage Induction Motors (DOL, Star-Delta, Auto-Transformer, Soft Starters).

Q28. Explain Permanent Magnet Synchronous Motors (PMSM), comparing Surface PMSM (SPMSM) with Interior PMSM (IPMSM).

Q29. Explain Transformer Protection: Differential Protection (Merz-Price scheme) and Buchholz Relay operation.

Q30. Analyze Symmetrical 3-Phase Short Circuit on an Alternator, deriving subtransient, transient, and steady-state reactances.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [A]
Q2: [A]
Q3: [A]
Q4: [A]
Q5: [A]
Q6: [A]
Q7: [A]
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Q98: [A]
Q99: [A]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

SECTION B: DETAILED SOLUTIONS FOR SUBJECTIVE QUESTIONS [illegible] leakage reactances a

3. Referring Secondary Parameters to Primary Side:
- Transformation ratio: K = N2 / N1.
- Secondary current referred to primary: I2' = K * I2.
- Secondary voltage referred to primary: V2' = V2 / K.
- To preserve identical I^2*R copper loss and I^2*X reactive VAR drops:
(I2')^2 * R2' = I2^2 * R2 => (K * I2)^2 * R2' = I2^2 * R2 => R2' = R2 / K^2.
Similarly: X2' = X2 / K^2.
4. Total Equivalent Parameters Referred to Primary:
- Equivalent resistance: R_eq1 = R1 + R2' = R1 + R2 / K^2.
- Equivalent leakage reactance: X_eq1 = X1 + X2' = X1 + X2 / K^2.
- Equivalent impedance: Z_eq1 = sqrt(R_eq1^2 + X_eq1^2).

[illegible]

% Regulation = [(V2_no_load - V2_full_load) / V2_full_load] * 100% = [(E2 - V2) / V2] * 100%.
2. Approximate Voltage Drop Derivation (Referred to Secondary):
- From secondary phasor diagram: E2 = V2 + I2 * R_eq2 * cos(phi) +- I2 * X_eq2 * sin(phi).
- Voltage drop: Delta-V = E2 - V2 approx I2 * R_eq2 * cos(phi) +- I2 * X_eq2 * sin(phi).
- Percentage Voltage Regulation:
% VR = [(I2 * R_eq2 * cos(phi) +- I2 * X_eq2 * sin(phi)) / V2] * 100%.
3. Analysis Across Load Power Factors:
- Lagging Power Factor (Inductive Load):
Sign is positive (+): % VR = [(I2*R_eq2*cos(phi) + I2*X_eq2*sin(phi)) / V2] * 100%.
Terminal voltage drops below no-load voltage. Regulation is positive.
- Leading Power Factor (Capacitive Load):
Sign is negative (-): % VR = [(I2*R_eq2*cos(phi) - I2*X_eq2*sin(phi)) / V2] * 100%.
If I2*X_eq2*sin(phi) > I2*R_eq2*cos(phi), regulation becomes negative (terminal voltage rises above no-load voltage).
- Zero Voltage Regulation Condition:
Occurs on leading power factor when numerator vanishes:
I2*R_eq2*cos(phi) = I2*X_eq2*sin(phi) => tan(phi) = R_eq2 / X_eq2 => cos(phi) = X_eq2 / Z_eq2 (leading).
- Maximum Voltage Regulation Condition:
Occurs on lagging power factor when d(VR)/d(phi) = 0 => tan(phi) = X_eq2 / R_eq2 (lagging).

>> Solution to Question 5:

[Problem Statement]: Derive the condition for Maximum Efficiency in a power transformer, and contrast power transformer design with distribution transforme

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[illegible]

2. Electromagnetic Torque Equation:
- Electric power converted to mechanical form: P_em = Eg * Ia.
- Mechanical power: P_mech = T_e * omega_m = T_e * (2 * pi * N / 60).
- Equating electric and mechanical power:
T_e * (2 * pi * N / 60) = Eg * Ia = [(Phi * Z * N * P) / (60 * A)] * Ia.
- Solving for electromagnetic torque T_e:
T_e = [(P * Z) / (2 * pi * A)] * Phi * Ia = 0.159 * (P * Z / A) * Phi * Ia Newton-meters.
- In short: T_e = k_t * Phi * Ia, where k_t is the machine design constant.

>>Solution to Question 8:
Problem Statement: Explain Armature Reaction in DC machines, its distorting and demagnetizing effects, and methods used for neutralization.
Compare DC Shunt, Series, and Compound Motors regarding torque-speed characteristics and industrial applications.

presses.
2. DC Series Motor:
- Circuit: Field winding connected in series with armature, carrying full armature current: Phi proportional to Ia (below saturation).
- Torque Relation: T proportional to Phi * Ia proportional to Ia^2 (quadratic starting torque).
- Speed Relation: N proportional to (V - Ia*(Ra + Rse)) / Phi approx 1 / Ia.
- Characteristic: At light load (Ia -> 0), Phi -> 0 and speed N skyrockets toward dangerous runaway levels. Must NEVER be run without mechanical load! High torque at low speeds.
- Applications: Electric locomotives (railway traction), cranes, hoists, trolley buses, automobile self-starters.
3. Cumulative Compound Motor:
- Circuit: Contains both series and shunt field windings assisting each other.
- Characteristic: High starting torque (from series field) combined with safe, stable no-load speed limiting (from shunt field).
- Applications: Heavy stamping presses, rolling mills, shears, reciprocating compressors, elevators.

[illegible]

1. Stator Spatial and Temporal Winding Architecture:
- Stator carries three identical phase windings (R, Y, B) physically displaced in space around the stator bore by 120 electrical degrees.
- Windings are energized by balanced 3-phase sinusoidal currents displaced in time by 120 electrical degrees:
i_R(t) = Im * sin(omega * t)
i_Y(t) = Im * sin(omega * t - 120 deg)
i_B(t) = Im * sin(omega * t - 240 deg)
2. Resulting Fluxes:
- Each phase current establishes a pulsating, stationary sinusoidal flux along its magnetic axis:
Phi_R = Phi_m * sin(omega * t)
Phi_Y = Phi_m * sin(omega * t - 120 deg)
Phi_B = Phi_m * sin(omega * t - 240 deg)
3. Mathematical Derivation of Net Resultant Flux (Phi_res):
- Resolve fluxes along orthogonal horizontal (X) and vertical (Y) axes:
Phi_x = Phi_R + Phi_Y * cos(120 deg) + Phi_B * cos(240 deg)
= Phi_m * [sin(wt) - 0.5 * sin(wt - 120) - 0.5 * sin(wt - 240)] = 1.5 * Phi_m * sin(omega * t).
Phi_y = 0 + Phi_Y * sin(120 deg) - Phi_B * sin(120 deg)
= Phi_m * (sqrt(3)/2) * [sin(wt - 120) - sin(wt - 240)] = - 1.5 * Phi_m * cos(omega * t).
- Net Flux Magnitude:
|Phi_res| = sqrt(Phi_x^2 + Phi_y^2) = 1.5 * Phi_m * sqrt(sin^2(wt) + cos^2(wt)) = 1.5 * Phi_m = Constant!
- Spatial Angular Orientation:
theta = arctan(Phi_y / Phi_x) = - omega * t.
4. Physical Conclusion:
- The resultant magnetic flux has a strictly constant magnitude equal to 1.5 times the maximum peak flux of an individual phase (1.5 * Phi_m).
- Its spatial orientation rotates uniformly in space at synchronous speed Ns = (120 * f) / P rpm.

>>Solution to Question 12:

[Problem Statement]: Derive the Torque-Slip Equation of a 3-phase Induction Motor, and analyze operating regions (motoring, generating, braking).

- Motoring Region (0 < s < 1, 0 < N < Ns):
Rotor rotates in direction of RMF below Ns. Electrical input converted to mechanical output torque.
- Generating Region (s < 0, N > Ns):
Rotor is driven above synchronous speed by external prime mover (e.g., wind turbine). Back-EMF reverses phase; machine converts mechanical input into electrical power delivered to grid.
- Braking / Plugging Region (s > 1, N is negative):
Stator phase sequence is suddenly reversed while rotor is spinning. Motor develops opposing counter-torque, rapidly arresting mechanical rotation.

[illegible]

= 1.0). Armature current Ia reaches absolute MINIMUM.
- Over-Excited (High If, Ef > V): Motor generates reactive power, supplying leading current to the grid (acts like a capacitor). Ia rises and leads voltage.
- Plotting Ia versus If produces characteristic 'V'-shaped curves for various load levels.

3. Inverted V-Curves (Power Factor cos phi vs If):
- Plotting cos phi versus If produces inverted 'V'-shaped curves with apex at unity power factor.

4. Synchronous Condenser Application:
- An over-excited synchronous motor operated with NO mechanical load connected to its shaft (P_shaft = 0).
- Draws near-zero active power while injecting large amounts of leading reactive power (kVAR) into electrical transmission substations to support grid voltage and correct lagging power factors.

[illegible] ratios.

2. Rotor Power Breakdown:
- In rotor equivalent circuit, rotor resistance R2/s is divided into:
R2 / s = R2 + R2 * [(1 - s) / s].
- The first term R2 represents physical Rotor Copper Loss:
P_rcu = 3 * I2^2 * R2 = s * P_ag.
- The second term R2 * (1 - s)/s represents Internal Gross Mechanical Power Developed:
P_mech = 3 * I2^2 * R2 * [(1 - s) / s] = (1 - s) * P_ag.
3. Fundamental Power Ratio:
P_ag : P_rcu : P_mech = 1 : s : (1 - s).
4. Shaft Useful Output Power:
- P_shaft = P_mech - (Mechanical Friction and Windage Losses + Stray Load Losses).
- Electromagnetic torque developed: T_e = P_ag / omega_s = P_mech / omega_r.

[illegible]

- From rated voltage point on ZPFC, draw horizontal line meeting OCC.
- Construct Potier triangle under curve:
* Vertical leg (DE) represents Armature Leakage Reactance Drop: I_a * X_potier.
* Horizontal leg (EF) represents Armature Reaction Equivalent Field Current: Ifa.
4. Determination of Voltage Regulation:
- Potier reactance: X_potier = DE / I_rated.
- Determine no-load excitation field current If0 by vectorially adding no-load field MMF (from OCC at voltage E = V + Ia*Ra + j*Ia*X_potier) and armature reaction MMF Ifa.
- From OCC, find Voc corresponding to If0.
- Voltage Regulation: % VR = [(Voc - V_rated) / V_rated] * 100%.

>> Solution to Question 20: Problem Statement: Explain [illegible] Field Oriented Control (Vector Control) of 3-Phase Induction Motors, deriving Clarke and Park transformations.

[i_alpha] = sqrt(2/3) * [1 -1/2 -1/2] [ia]
[i_beta] [0 sqrt(3)/2 -sqrt(3)/2] [ib]
[ic].
- Park Transformation (2-phase stationary alpha-beta to rotating dq-reference frame aligned with rotor flux vector at angle theta_e):
[id] = [cos(theta_e) sin(theta_e)] [i_alpha]
[iq] [-sin(theta_e) cos(theta_e)] [i_beta].
3. Decoupled Control:
- Direct-axis current (ids): Controls rotor flux linkage Psi_r exclusively (equivalent to DC field current If).
- Quadrature-axis current (iqs): Controls electromagnetic torque T_e exclusively (equivalent to DC armature current Ia).
- Torque Equation: T_e = (3/2) * (P/2) * (Lm / Lr) * Psi_r * iqs.
- Allows instant, independent four-quadrant torque and speed dynamic response.

[illegible]

Phi_max = 2 * Phi_m + Phi_r (can reach ~2.5 to 2.8 times normal rated flux).
2. Consequences:
- Transformer core enters extreme magnetic saturation (permeability collapses toward air).
- Magnetizing inrush current spikes to 10 to 20 times rated full-load current.
3. Protection Strategy:
- Inrush current contains extraordinarily high Second Harmonic content (~20-40% of fundamental).
- Differential protection relays incorporate Second-Harmonic Restraint filters that block relay tripping during energization while allowing tripping on genuine internal faults.

[illegible]

- Sensor signals generate a 6-state digital switching sequence (Trapezoidal / Six-Step Commutation).
- At any instant, two phases conduct current (one high-side switch, one low-side switch), while the third phase remains unenergized.
- The back-EMF waveform is trapezoidal, producing flat-topped torque when energized with rectangular block currents.

[illegible]

[illegible]

[illegible] state reactances.
